[INCH-POUND]
KKK-S-2922
April 21, 1997
SUPERSEDING
MIL-S-45344H
4 December 1991

FEDERAL SPECIFICATION

SEMITRAILER, VAN; COMMERCIAL (FURNITURE, PERSONNEL, REFRIGERATOR, CARGO, AND SIDE CURTAIN) 7 TO 35 TON

The General Services Administration has authorized the use of this specification by all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers furniture, personnel, refrigerator, closed van, open van, and side curtain semitrailers, having two or four dual wheels with payload capacities of 14,000 to 70,000 pounds (lb) (6 342 to 31 710 kilogram (kg)).

1.2 Classification. Semitrailers shall be one of the following types, classes, and sizes, as specified (see 6.2). Class of semitrailer shall be determined by the payload rating. Size of the semitrailer shall be determined by the overall length dimension. Applicability of classification shall be as shown in table I.

Type I	-	Semitrailer, furniture.
Type II	-	Semitrailer, personnel.
Type III	-	Semitrailer, refrigerator.
Type IV	-	Semitrailer, closed van.
Type V	-	Semitrailer, open van.
Type VI	-	Semitrailer, side curtain van.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 15E2), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2330

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

Class A - 14,000 lb (6 342 kg).
Class B - 18,000 lb (8 154 kg).
Class C - 24,000 lb (10 872 kg).
Class D - 40,000 lb (18 120 kg).
Class E - 50,000 lb (22 650 kg).
Class F - 70,000 lb (31 710 kg).

Size 28 - 28-foot length (8 534 millimetre (mm)).
Size 30 - 30-foot length (9 144 mm).
Size 32 - 32-foot length (9 754 mm).
Size 35 - 35-foot length (10 668 mm).
Size 40 - 40-foot length (12 192 mm).
Size 48 - 48-foot length (14 630 mm).

TABLE I. Classification applicability.

Type	Class	Size					
		28	30	32	35	40	48
I	C	-	X	-	X	X	
II	A	X	-	-	-	-	
	B	-	-	-	X	-	
III	A	X	-	-	-	-	
	C	-	X	-	-	-	
	D	-	-	X	X	X	
IV	C	-	X	-	-	-	
	D	-	-	X	X	X	
	F	-	-	-	-	-	X
V	C	-	X	-	-	-	
	D	-	-	X	X	X	
VI	E	-	-	-	-	-	X

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard

FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.

Commercial Item Descriptions

A-A-393 - Extinguisher, Fire, Dry Chemical (Hand-Portable).
A-A-50271 - Plates, Identification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

A-A-52520 - Hardwood; Floorboards and Platforms, for Military Vehicles.

Military Specifications

MIL-PRF-25959 - Tie Downs, Cargo, Aircraft.

Military Standards

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MS75021 - Connector, Receptacle, Electrical 12-Contact, Intervehicular, 28 Volt (V), Waterproof.

(Copies of federal and military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the Defense Automated Printing Services, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.1 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Standards and Regulations.
Federal Motor Vehicle Safety Regulations.

(Applications for copies should be addressed be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

THE EUROPEAN TYRE AND RIM TECHNICAL ORGANIZATION (ETRTO)

Standard Manual.

(Private sector and civil agencies may purchase copies of this voluntary standard from the European Tyre and Rim Technical Organization, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA - MG 1 - Motors and Generators.

(Private sector and civil agencies may purchase copies of this voluntary standard from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electric Code

(Private sector and civil agencies may purchase copies of this voluntary standard from the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J318	-	Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers Trucks, Trucks-Tractors, and Trailers.
SAE J534	-	Lubrication Fittings.
SAE J551	-	Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
SAE J560	-	Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
SAE J682	-	Rear Wheel Splash and Stone Throw Protection.
SAE J700	-	Upper Coupler Kingpin - Commercial Trailers and Semitrailers.
SAE J702	-	Brake and Electrical Connection Locations - Truck-Tractor and Truck-Trailer.
SAE J1292	-	Automobile, Truck, Truck-Tractor, Trailer, and Motor Coach Wiring.

(Private sector and civil agencies may purchase copies of this voluntary standard from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Private sector and civil agencies may purchase copies of this voluntary standard from the Tire and Rim Association, Inc., 175 Montrose West Avenue, Copley, OH 44321.)

(DoD activities may obtain copies of those adopted voluntary standards listed in the DoD Index of Specifications and Standards free of charge from the Defense Automated Printing Services, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First production vehicle. The contractor shall furnish a semitrailer for first production vehicle inspection.

3.2 Standard semitrailer, components, and accessories. Except as specified in 3.2.1 through 3.2.1.8, the semitrailer components and accessories, shall be standard or optional items which meet or exceed the requirements of this specification. All items shall be as represented and rated in the manufacturer's sales information, including special or mounted equipment. Sales information shall be limited to specifications and technical material, identical to that furnished to the authorized company representative for selection of semitrailer models and components, and shall be on file in the engineering offices of the procuring activity prior to delivery of the items. Semitrailer, components, assemblies, and accessories to be provided hereunder are standard or optional commercial products for the purpose of this specification. Such items shall be construed as commercial motor vehicle products, and shall comply with all DoT Federal Motor Vehicles Safety Standards and Regulations applicable, on the date of manufacture.

3.2.1 Special requirements. In addition to standard semitrailers and components specified in 3.2, semitrailers shall be furnished with special equipment as specified herein.

3.2.1.1 Painting and marking. Treatment, painting, marking, and data plates shall be in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color and markings shall be in accordance with MIL-STD-1223.

3.2.1.2 Rustproofing. When specified (see 6.2), the semitrailer shall be rustproofed in accordance with FED-STD-297.

3.2.1.3 Hardwood. The platform shall be floored with apitong, white or red oak, or white ash as specified in A-A-52520. Apitong is allowed for flooring only, provided the average weight shall be not less that 50 lb per cubic foot (ft^3) (801 kilogram per cubic metre (kg/m^3)), with none of the individual pieces weighing less than 47 ft^3 (753 kg/m^3), with no more than 15 percent moisture content at time of weighing. Other wood parts specified herein to be furnished in hardwood shall be fabricated from the types of hardwood specified in A-A-52520.

3.2.1.4 Wood treatment. Wood surfaces shall be treated in accordance with the requirements of MIL-STD-1223. Apitong need not be treated.

3.2.1.5 Radio interference suppression. The semitrailer shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

3.2.1.6 Lifting and tiedown attachments. When specified (see 6.2), the semitrailer shall be equipped with lifting and tying down attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A transportation plate conforming to composition A (class 1 or 2) or composition C of A-A-50271 shall be attached to the semitrailer. The transportation plate shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the semitrailer, showing the center of gravity shall be provided on the transportation plate. Tiedown attachments shall be identified by stenciling or other suitable markings. Tiedown markings shall clearly indicate that the attachments are intended for the tiedown of the semitrailer on the carrier when shipped.

3.2.1.7 Ratings. Semitrailer rating shall be the manufacturer's published rating. Component and semitrailer ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the engineering offices of the procuring activity. Minimum payload shall conform to 1.2 for the specified type, class, and size of the semitrailer.

3.2.1.8 Cargo tiedown rings. When specified (see 6.2), cargo tiedown rings shall be furnished. Tiedown rings shall be flush mounted in the floor and shall accept the MB-1 tiedown devices as specified in MIL-PRF-25959. Tiedown ring installation shall use existing semitrailer structure to the maximum extent possible. Each tiedown ring and its attachment shall be capable of withstanding not less than 2,000-lb (906 kg) pull in the vertical direction and at a 45-degree angle to the vertical direction. Tiedown rings shall be corrosion-resistant. Tiedown rings shall be located in tow longitudinal rows, 24 inches (610 mm) on either side of the semitrailer centerline. Each row shall consist of six or more tiedown rings, symmetrically spaced with not more than 36 inches (914 mm) longitudinal distance between rings.

3.2.1.9 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

3.3 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.4 Weights, loads, and dimensions.

3.4.1 Net weight. The net weight of the fully equipped semitrailer shall include the chassis weight and the weight of all attachments, accessories, and equipment and shall be the minimum practicable for the service intended.

3.4.2 Rated payload capacity. The rated payload capacity, evenly distributed over the load space, shall be not less than specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

TABLE II. Payloads and clearance dimensions.

Type	Class	Payload rating (not less than) (pounds) (kilograms)	Kingpin center to front of semitrailer (not less than) (inches) (millimetre)	Height floor to ground at rear (not more than) (inches) (millimetre)	Swing radius (not more than) (inches) (millimetre)
I	C	24,000 (10 872 kg)	24 (610 mm)	31 (787 mm)	52.5 (1 334 mm)
II	A	14,000 (6 342 kg)	24 (610 mm)	32 (813 mm)	58.0 (1 472 mm)
	B	18,000 (8 154 kg)	24 (610 mm)	32 (813 mm)	58.0 (1 472 mm)
III	A	14,000 (6 342 kg)	24 (610 mm)	58 (1 472 mm)	52.5 (1 334 mm)
	C	18,000 (8 154 kg)	24 (610 mm)	58 (1 472 mm)	52.5 (1 334 mm)
	D	14,000 (6 342 kg)	36 (914 mm)	58 (1 472 mm)	61.5 (1 562 mm)
IV	C	24,000 (10 872 kg)	24 (610 mm)	54 (1 372 mm)	52.5 (1 334 mm)
	D	40,000 (18 120 kg)	36 (914 mm)	54 (1 372 mm)	58.0 (1 472 mm)
	F	70,000 (31 710 kg)	36 (914 mm)	54 (1 372 mm)	60.0 (1 524 mm)
V	C	24,000 (10 872 kg)	24 (610 mm)	54 (1 372 mm)	52.5 (1 334 mm)
	D	40,000 (18 120 kg)	36 (914 mm)	54 (1 372 mm)	58.0 (1 472 mm)
VI	E	50,000 (22 650 kg)	36 (914 mm)	54 (1 372 mm)	58.0 (1 472 mm)

3.4.3 Gross weight. The gross vehicle weight (gvw) shall consist of the net weight and the rated payload evenly distributed over the load space.

3.4.4 Dimensions. Dimensions and clearances, with the semitrailer without payload, shall conform to table II and the following:

a. Overall height, not more than 150 inches (3 810 mm).

b. Overall width, not more than 96 inches (2 438 mm).

c. Ground clearance, not less than 9 inches (229 mm).

d. Landing gear ground clearance, see 3.6.7.

e. Minimum swing clearance (radius from centerline of kingpin to any portion of semitrailer 6 inches (152 mm) or more below upper fifth wheel plate) 82 inches (2 083 mm).

f. Upper fifth wheel height 48 inches $\pm$ 1 inch (1 219 mm $\pm$ 25 mm).

3.4.5 Wheel loading. The axle(s) of the semitrailer shall be positioned so that the proportion of the gvw supported on the tires, measured at the ground with the rated payload evenly distributed, shall not impose a total wheel loading in excess of 19,000 lb (8 607 kg) for single axle semitrailers or 34,000 lb (15 402 kg) for tandem axle semitrailers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

3.5 Performance. The semitrailer, fully equipped and loaded with rated payload, shall show no evidence of part failure or permanent deformation when towed at speeds as great as 10 miles per hour (mph) (16 kilometre per hour (km/h)) over unimproved roads (see 6.4.2), and over reasonably hard, uneven terrain; and when towed at speeds as great as 55 mph (88 km/h) over improved roads (see 6.4.1). The empty semitrailer shall show no part failure or permanent set when towed at these speed and terrain combinations.

3.5.1 Turning ability. The semitrailer shall be capable of assuming a 90-degree angle to the coupled towing vehicle without cramping or damage to the semitrailer or the towing vehicle.

3.5.2 Tracking ability. The semitrailer shall conform to the tracking requirements of DoT Federal Motor Carrier Safety Regulations, section 393.70(a).

3.5.3 Brake performance. Service brakes shall comply with the performance requirements of DoT Federal Motor Carrier Safety Regulations, section 393.52.

3.6 Chassis components.

3.6.1 Suspension system. The semitrailer shall be furnished with the manufacturer's standard suspension system. Each component of the suspension system shall have a rated capacity at least equal to the load imposed, measured at the ground, when the semitrailer is loaded with its rated payload (see 3.4.2). Clearances shall preclude interference between tires and any other part of the semitrailer under the operating conditions specified herein.

3.6.1.1 Air suspension system. When specified (see 6.2), the semitrailer shall be equipped with an air suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground with the semitrailer loaded as specified in 3.4.2. Hydraulic double-acting shock absorbers shall be provided on the axles. The suspension system shall incorporate leveling valves with time delays to avoid constant air consumption. The suspension system shall have none of the sprung weight carried on anything other than the air springs. No other spring medium, such as leaf, coil, or rubber springs, may be used in conjunction with the air springs. The air suspension system shall incorporate radius rods to control lateral and longitudinal movement. Radius rods shall transmit pulling and braking forces to the chassis and body. Each end of the radius rods shall be equipped with rubber bushings that do not require periodic lubrication. The suspension shall be provided with a mechanism at each wheel to ensure lifting of the wheel and axle when jacking the semitrailer from the applicable jacking location.

3.6.2 Axle(s). Class A, B, and C semitrailers shall be furnished with a single axle. Class D, E, and F semitrailers shall be furnished with tandem axles. Type VI tandem axles shall have sliding capabilities. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground with semitrailer loaded with rated payload. The wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. The hubcaps shall have a window for visual determination of oil level. The hubs shall be fitted with seals. Provision for venting, or other method of withstanding internal pressure buildup, and for replenishing the oil supply shall be incorporated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

3.6.3 Wheels, rims, and tires.

3.6.3.1 Wheels, rims, and tires. Semitrailer shall be equipped with dual wheels on each axle. When specified (see 6.2), disk type wheels shall be furnished. The tires shall conform TRA or ETRTO recommendations. Tire and rim sizes shall be the same for all wheels of the semitrailer. Tire shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with semitrailer loaded with rated payload. When specified (see 6.2), semitrailer shall be equipped with a wheel mounted mile meter. Tires shall be of the tubeless type. Multi-piece rims shall not be furnished. Rims shall be painted the same color as the trailer.

3.6.3.2 Tire carrier. When specified (see 6.2), a spare tire carrier shall be provided and installed in a readily accessible location under the vehicle. Carrier shall be located forward of the axle(s) on the curbside. Means shall be provided for securing the tire within the carrier to prevent accidental loss. Threaded fasteners when used to secure the spare tire in the carrier, shall be constructed of, or plated with, corrosion-resistant material.

3.6.3.3 Spare wheel. When specified (see 6.2), a spare wheel or rim shall be mounted on the tire carrier.

3.6.3.4 Spare tire. When specified (see 6.2), an inflated spare tire shall be mounted on the spare wheel or rim specified in 3.6.3.3. The spare tire shall be of the same manufacture, size, ply rating, and tread design as the tires installed on the semitrailer.

3.6.4 Brakes.

3.6.4.1 Service brakes. Service brakes shall be of the full air type and shall conform to Federal Motor Carrier Safety Regulations 393.43 through 393.47. The braking system shall include automatic slack adjusters, piping, hose connections, gladhands, spring loaded dust covers or dummy gladhands equipped with security chains, and other components required for a complete air brake system. Brake linings shall be of nonasbestos material. Gladhands shall conform to SAE J318. Air hose location shall comply with SAE J702. The braking system shall be installed in a manner which provides road clearance for travel over uneven terrain and protection against damage caused by objects striking components. No part of the braking system shall extend below the bottom of wheel rims.

3.6.4.2 Parking brakes. Spring or air diaphragm mechanical lock type parking brakes shall be provided. The parking brakes shall be automatically applied upon disconnection of the emergency air line and under emergency braking conditions. The parking brakes shall remain in the applied conditions with no additional application and despite the depletion of all air pressure. Parking brakes shall conform to Federal Motor Carriers Safety Regulation 393.41. With no additional application, the parking brakes shall hold the semitrailer, with rated payload, on a 10 percent grade despite the depletion of the compressed air supply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

3.6.5 Upper fifth wheel plate. The upper fifth wheel plate shall be designed for coupling to a full oscillating and fore and aft rocking fifth wheel, and shall be of sufficient strength to support a fifth wheel 36 inches (914 mm) in diameter. In addition, the coupling device shall conform to DoT Federal Motor Carrier Safety Regulation, section 393.70. The kingpin shall be of heat-treated alloy steel and shall conform to SAE J700. The forward end of the upper fifth wheel plate shall have a turned-up lip not less than 2 inches (51 mm) high for ease of coupling with the towing vehicle.

3.6.6 Lighting. All lights and reflectors shall be protected from operational hazard by mounting in recessed or otherwise guarded locations. The brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Lights and reflectors shall not be mounted on vertical surfaces of rub rails (unless recessed and fully protected) or on the semitrailer bumper. The design and construction of the lamps shall be in accordance with DoT Federal Motor Vehicle Safety Standards and Regulations 571.108. Requirements and placement of lamps, reflectors, and electrical wiring shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.25 and section 393.31. All electrical wiring shall conform to SAE J1292. Clearance and identification lights shall be constructed for easy removal and replacement of lamps and lenses.

3.6.6.1 Receptacle, 12V direct current. The front of the semitrailer shall be equipped with a 7-contact receptacle conforming to SAE J560 with the connectors connected and color coded as specified therein. The receptacle shall be provided with a spring loaded cover. The receptacle shall be located in accordance with SAE J702.

3.6.6.2 Twelve-volt direct current system. A 12V direct current lighting system conforming to 3.6.6 and 3.6.6.1 shall be provided. The 12V lighting system shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.14, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32 and 393.33.

3.6.6.3 Independent 24V system. When specified (see 6.2), a 24V lighting system independent of the 12V lighting system shall be provided. Separate wiring and bulbs shall be provided for the 24V system. The system shall conform to 3.6.6. The 24V system shall also conform to the DoT Federal Motor Carrier Safety Regulations referenced in 3.6.6.2, except the 24V stop and turn lamps shall be combined.

3.6.6.4 Interconnected 24V system. When specified (see 6.2), capability of operating the 12V system (see 3.6.6.2) from a towing vehicle equipped with a nominal 24V electrical system shall also be provided.

3.6.6.5 Receptacle, 24V. When either an independent or interconnected 24V system is specified, the front of the semitrailer shall also be equipped with a 12-contact receptacle and cover conforming to MS75021, part number MS75021-1. The receptacle shall be located in accordance with SAE J702. The receptacle shall be connected to the lights as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

Contact B	Connect to left-hand turn signal and stop lamp (yellow).
Contact D	Connect to ground (white).
Contact E	Connect to clearance, side marker, identification, and tail lamps (black and brown).
Contact J	Connect to right-hand turn signal and stop lamp (green).
Contact L	Connect to ground (white).

The rest of the contact shall not be connected. Circuits B and J on tactical trucks are combination stop and turn indicator circuits. On the interconnected 24V system, the normal 12V turn signal lights will function both as turn signals and stop lights, and the normal 12V stop lights will not be operational when the semitrailer is connected to a towing vehicle with a 24V power supply. Because of this condition, the stop light (red) circuit is not connected to the 24V 12-contact receptacle.

3.6.6.6 Resistors. When an interconnected 24V system is specified (see 3.6.6.4), the 12-contact receptacle shall be provided with resistance in each circuit to reduce the voltage of the tactical (military design) towing vehicle from a nominal 24V to 12V. Each circuit resistor shall be selected to reduce the 28V regulated voltage of the tactical truck to within the maximum rated voltage of the semitrailer electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a heat sink to preclude overheating and any damage to resistors, wiring or adjacent components. Manufacturer's solid state assembly is acceptable in lieu of resistor assembly.

3.6.7 Landing gear. Semitrailer shall have two vertical lift, telescopic, nonrotating landing legs with two speed gears with connecting cross-shaft and a handcrank on the curbside. Landing legs shall be equipped with steel wheels or self-leveling skid pads. Supports for the crank extension shafts and clips for holding cranks when folded shall be provided. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to proportion of gross weight sustained and the forces resulting from impact during coupling and uncoupling operations. The landing gear assembly shall be mounted to mounting plates that are gusseted to resist forward and rearward thrust on the landing legs. When placed in travel position, the landing gear legs shall remain positively locked. The landing gear shall be protected to preclude the entrance of foreign matter which would impair its functioning or mechanical efficiency. The landing gear shall have a range of adjustments to vary the height of the upper fifth wheel plate, at centerline of kingpin, from 47 inches (1 694 mm) to at least 52 inches (1 321 mm) from the ground. With the semitrailer coupled to a towing tractor or support to simulate the fifth wheel height specified in 3.4.4(f), the clearance under the fully retracted landing gear shall exceed the semitrailer ground clearance, but shall in no case be less than 12 inches (305 mm).

3.6.8 Rear end protection. Semitrailer rear end protection shall be in accordance with DoT Federal Carrier Safety Regulations, section 393.86. For type II semitrailers, a rear step bumper as specified in 3.6.8.1 shall be furnished. For type III, IV, V, and VI semitrailers, a dock bumper and an extension step bumper as specified in 3.6.8.2 and 3.6.8.3 shall be furnished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

3.6.8.1 Rear step bumper. A rear step bumper shall be furnished and located approximately 8 inches (203 mm) below the floor level of the type II semitrailer. The step bumper shall extend full width of the semitrailer and not less than 8 inches (203) to the rear of the body. The top of the rear step bumper shall be covered with not less than 0.1196-inch (3 mm) thick nonskid sheet steel.

3.6.8.2 Dock bumper. A dock bumper, extending not less than 1.5 inches (38 mm) beyond the rear doors shall be furnished. The dock bumper shall be full width of vehicle and shall be located approximately at floor level.

3.6.8.3 Extension bumper. An extension type step bumper shall be furnished consisting of a horizontal bumper (step), vertical extensions, and angular extension supports. Bumper extension supports shall be attached to the bumper (step) and semitrailer frame members. The extension bumper shall be of all welded construction and be fabricated from steel members. The rear edge of the bumper (step) shall not protrude to the rear of the dock bumper and shall be located not more than 1.5 inches (38 mm) forward of the rear edge of the dock bumper. The bumper step height shall be approximately half-way between ground level and floor level.

3.6.9 Rear wheel splash and stone throw protection. Rear wheels shall have mud flaps at rear. A metal strip not less than 0.125-inch (3 mm) thick and not less than 1-inch (25 mm) wide, extending the entire width of the mud flap, shall be installed to prevent the bold heads or bolt nuts from damaging the mud flaps. As an alternate method of attaching the mud flaps, taps or clips with surface contact dimensions of not less than 1-inch (25 mm) high by 1.25-inch (32 mm) wide by 0.09375-inch (2 mm) thick shall be furnished at each bolt. All splash shield and mud flap installations, front and rear, shall conform to the rear wheel splash and stone throw protection provisions of SAE J682. Mud flaps shall be of the plain black style.

3.7 Construction. Unless otherwise specified (see 6.2), body construction shall be stressed-skin (monocoque or semi-monocoque). Unless otherwise specified (see 6.2), exterior skin, consisting of roof, sides, and end sheets utilized in body construction, shall be aluminum alloy. When exterior sheets are furnished without formed corrugations, the sheets shall be stretcher-leveled. Materials shall be free from any defects or imperfections that may affect the performance and appearance of the vehicle. Aluminum alloys shall have corrosion-resistant properties inherent in the chemical composition of aluminum. Either interior or exterior side wall posts shall be utilized. Body structural members may be steel or aluminum alloy except rear body members shall be steel. Parts of dissimilar metals jointed together shall be protected on contacting faces to prevent galvanic action. Doors, body, roof, and panel joints shall be watertight and leakproof. All doors, windows, and other cutouts shall be reinforced.

3.7.1 Roof panels. Roof shall be of one piece and of stretcher level aluminum with roof bows on not less than 24-inch (610 mm) centers. A drip molding or rain gutter shall be provided over the doors. The roof shall be secured to roof bows to prevent panel vibration. The roof shall be capable of withstanding a static load of 30 lb (14 kg) on any square foot of area without damage. The roof shall be crowned to permit drainage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

3.7.2 Front end. The front end shall have beveled or rounded corners which sacrifice a minimum of the interior cube. Front end shall conform to the swing radius requirement of table II.

3.8 Semitrailer types.

3.8.1 Type I (furniture). Furniture semitrailer shall be fully enclosed, and provided with double rear doors. Semitrailer shall have payload capacity and clearance dimensions conforming to 3.4.2 and 3.4.4. The overall length dimension, including door hardware and rear bumper, shall be as indicated by the applicable size. The inside body dimensions shall be as follows:

a. Width, not less than 92 inches (2 337 mm).

b. Height, ahead of drop, not less than 94 inches (2 388 mm).

c. Height, behind drop, not less than 115 inches (2 921 mm).

3.8.1.1 Type I body interior. Interior panels shall be attached to the wall and ceiling members. Panels shall be of exterior grade AD plywood not less than 0.25-inch thick. Interior radius corners may be aluminum lined. Not less than four rope cleats or rope tiepins shall be provided on each sidewall post back of the drop and not less than three on each post forward of the drop. Cleats or tiepins shall be recessed and shall not protrude beyond the lining. Rug rods or straps shall be furnished in accordance with the manufacturer's standard practice. Not less than two 12V dome lights with control switches at each door opening shall be provided.

3.8.1.2 Type I platform. The platform shall be of the drop type. Rub rails shall extend the full length of the semitrailer. Floor shall be not less than 1.125-inch (29 mm) thick hardwood or not less than 1.0625-inch (27 mm) thick laminated hardwood, in length of not less than 6 feet (1 829 mm). Boards shall be laid lengthwise with a tongue-and-groove or shiplap joints. Flooring shall be secured with countersunk zinc-coated or phosphate-coated screws or bolts. Not less than two securing screws or bolts shall be installed at each cross sill, and not less than two on each side of each butt joint. Top of countersunk screw or bolt head shall be not less than 0.125-inch (3 mm) and not more than 0.25-inch (6 mm) below top of wood surface. Butt joints shall be centered over crossmembers and staggered. Rails or pockets for attaching loading ramps (see 3.8.1.5), shall be provided at the rear and the side door location of the semitrailer platform.

3.8.1.3 Sidewall and roof framing. Sidewall framing shall have pressed steel posts, welded or riveted to the body frame. Outside sidewall frame members shall be full length. Sidewall posts and roof bows shall be spaced on not more than 24-inch (610 mm) centers.

3.8.1.4 Type I doors. Doors shall be hung between box posts with not less than three heavy-duty pin strap hinges on each door permitting the doors to fold back against the sides of the body. Doors shall be provided with double cam latches and operating handle. Rear door opening shall be not less than 107 inches (2 718 mm) high by 88 inches (2 235 mm) wide. Double curbside doors, covering a door opening not less than 84 inches (2 134 mm) wide and not less than

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

96 inches (2 438 mm) high, shall be furnished. When specified (see 6.2), sliding type curbside door(s) shall be furnished. Means shall be provided to hold all doors in the full open position.

3.8.1.5 Loading ramps. Two loading ramps of all metal construction, complete with hooks and handles, shall be furnished. Each ramp shall be approximately 40 inches (1 016 mm) wide with a ramping angle of not less than 12 degrees and not more than 15 degrees. Each ramp shall be capable of rigidly supporting a concentrated load of 1,000 lb (453 kg) without permanent deformation.

3.8.1.6 Roll-up rear door. When specified (see 6.2), a roll-up overhead rear door shall be provided. The door shall provide a clear opening of not less than 88 inches (2 235 mm) in width and not less than 107 inches (2 718 mm) in height. The door shall be of the sectional type having not less than six sections. The door sections shall incorporate joints of the tapered tongue and groove or shiplap type. The door shall be plastic-coated or aluminum-covered 0.75-inch (19 mm) marine plywood construction. Door section corrosion-resistant hinges shall be mounted on the inside of the door, one on each side of each section joint. The door track shall be of the corrosion-resistant construction type equipped with a positive stop at the end of the track. Door rollers, counter-balance unit, and cables of corrosion-resistant construction shall be furnished. The door shall be weather tight and shall close to not less than 0.25-inch (6 mm) below floor level. A heavy-duty corrosion-resistant cam-operated lever-type lock shall be provided, equipped with provision for use of a padlock. One heavy-duty corrosion-resistant closed type grab handle shall be provided on the left outside bottom of the door. One nylon double loop pull down strap not less than 2 inches (51 mm) wide and 12 inches (305 mm) in length shall be provided and located adjacent to the door lock.

3.8.1.7 Roll-up side door. When specified (see 6.2), a roll-up overhead type door, located on the curbside (right) of the semitrailer shall be furnished. The door shall provide a clear opening of not less than 84 inches (2 134 mm) in width and not less than 96 inches (2 438 mm) in height. Otherwise, the door shall conform to 3.8.1.6.

3.8.2 Type II (personnel). Passenger capacity, including standees, shall be 80 for class A and 100 for class B. Personnel semitrailer shall be fully enclosed, rainproof, provided with entrance door(s), emergency exit, windows, seating, stanchions, and grabrails. The type II personnel semitrailer shall meet DoT Federal Carrier Safety Regulations for the term “Bus” as defined in DoT Federal Carrier Safety Regulations, section 390.3 and section 398.1. Unit shall further comply with applicable DoT Federal Motor Vehicle Safety Standards identified in this specification and other detailed DoT Federal Motor Carrier Safety Regulations of sections 393 and 398.5 as specified herein. Semitrailer shall have payload capacity and clearance dimensions conforming to 3.4.2 and 3.4.4. The overall length dimension, exclusive of rear bumper, shall be as indicated by the applicable size. The inside body dimensions shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

a. Width, not less than 90 inches (2 286 mm).

b. Height, rear section, not less than 100 inches (2 540 mm).

c. Height of kickup, 18 inches (457 mm) to 26 inches (660 mm).

3.8.2.1 Sidewall frames. The metal outer rails of body frame shall be continuous without splice, except at stepwells, over wheels, and at the kickup and shall be supported by metal cross sills extending the full width of the body. Spacing of cross sills shall be not greater than 30 inches (762 mm).

3.8.2.2 Type II body interior. The body interior (panels) except the floor, shall be constructed of 0.063 aluminum alloy sheet material. The lining shall be secured to each body frame member with fasteners spaced not more than 4 inches (102 mm) apart.

3.8.2.3 Type II platform. The platform shall be of the drop type. Floor material and construction shall be as specified in 3.8.1.2 for type I semitrailer. A step shall be provided to the kickup platform on centerlines of each aisle. Step shall be not less than 21 inches (533 mm) wide with 10-inch (254 mm) tread and height equal to half the kickup height.

3.8.2.4 Type II insulation. The body wall and roof insulation shall provide resistance to transfer of heat equivalent to the protection afforded by a 1-inch (25 mm) thick batt insulator, having a thermal conductivity of 0.24 British thermal units (Btu) per square foot per degree Fahrenheit (°F) per inch (0.04 watt per metre kelvin (W/(m·K) per 25 mm) of thickness at a mean temperature of 70 °F (21 degrees Celsius (°C)). Insulation shall be cellular material, nonhygroscopic, fungus resistant, and fireproof. In addition, a spray coat of sound deadening or insulating material shall be applied to the inner (not exposed) surfaces of panels and other sheet metal as added insulation and to prevent drumming.

3.8.2.5 Entrance. For class A and B semitrailers, an entrance shall be provided on the right side of the semitrailer, beginning not less than 10 inches (254 mm) or more than 2 feet (610 mm) to the rear of the kickup. For class B, in addition to the entrance at the kickup, a second entrance shall be provided on the right side not less than 10 inches (254 mm) or more than 2 feet (610 mm) forward of the wheel well. Entrances shall have double, bus type, air actuated doors which provide not less than a 43-inch (1 092 mm) clear opening with the doors in the open position. Entrance doors shall be equipped with the manufacturer's standard sizes and quantity of glass windows. Doors shall be provided with effective compression or overlapping seals as required to prevent injury to passengers and entry of dust, rain, or cold air. Rubber weather stripping shall be installed at the bottom of each door. The top of the door opening shall be not less than 77 inches (1 956 mm) above the semitrailer floor. A rain gutter shall be installed on the outside over the entrance doors. A step well shall be provided. The step well shall be of metal construction with two steps. The first step at the kickup door shall have a tread depth of not less than 10 inches (254 mm) and not more than 15 inches (381 mm). The second step shall have not less than 10 inches (254 mm) tread depth and shall extend from the center to the rear of the entrance opening. The rear side door (class B) shall have full width second step. The distance from

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

ground level to the first step shall be not greater than 15.5 inches (394 mm) with semitrailer unloaded. The tread shall be constructed from 0.1875-inch (5 mm) steel plate. Piping shall be provided to allow operation of the doors with air supplied by the towing vehicle. Individual controls for operation of entrance doors shall be suitably located to the rear of each door. An access covered by hinged, weatherproof plate or a rod (push-pull or hinged) connected to the door controls, extending to the outside and weatherproofed shall be furnished at the side entrance to permit operation of the door control from the outside of the semitrailer. The access plate shall be furnished with adequate latches, operable from outside the semitrailer, to secure it in the closed position during travel. Controls shall be marked to identify their function. A 5-inch (127 mm) diameter red light, that will flash continuously when the side door or emergency exit door is open, shall be furnished and installed on the front of the semitrailer near the curbside, not less than 7 feet (2 134 mm) from ground level. In addition, to prevent accidental application of the door operating control, the control shall be covered with a hinged-type, spring-held, normally closed cover. Floor and step treads shall be coated with the manufacturer's standard nonslip, abrasive filled, synthetic binder compound.

3.8.2.6 Emergency exit. A centrally located emergency exit not less than 47 inches (1 194 mm) wide and not less than 77 inches (1 956 mm) high shall be provided in the rear of the semitrailer. The exit shall be equipped with two reinforced side hung doors of metal construction with 0.25-inch (6 mm) plywood lining of aluminum with plywood core and extruded aluminum trim. A rain gutter shall be installed on the outside over the emergency exit doors. The doors shall be provided with weather stripping, and shall be secured in the closed position by a two point latch mechanism, operable from both inside and outside the semitrailer. The door latch mechanism shall be designed to prevent accidental opening. Marking and lighting of emergency exits shall be provided on all type II semitrailers to conform to DoT Federal Motor Carrier Safety Regulations, section 393.92.

3.8.2.7 Window openings. Window openings shall be provided in alternate spaces between upright body posts, on each side of the body, for the entire length of the semitrailer. The openings shall be approximately 54 inches (1 372 mm) from main floor line to sill, and shall be the width of the space between posts, and not less than 22 inches (559 mm) high. Windows of a standard commercial sliding sash bus type, with an operating mechanism for opening and closing, shall be installed in the window opening. Windows shall be rattleproof and when closed, rainproof and windproof. Rain gutters shall be installed on the outside over all windows that open. All regular side windows shall be the push out type conforming to DoT Federal Motor Carrier Safety Regulations, sections 393.61(b) and (c), 393.62, 393.63, and 571.217, and shall be not less than 17.75 inches (451 mm) wide. Glass for all windows and doors shall be of laminated safety type, and shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.60(a). When specified (see 6.2), the outside of the window opening shall be covered with an expanded aluminum flattened industrial mesh of not less than 0.0598-inch (1.5 mm) thickness. The mesh shall not interface with opening and closing the windows.

3.8.2.8 Seating arrangements. Slatted oakwood benches, 18 inches (457 mm) from the floor, shall be furnished and installed in the semitrailer body. The benches installed against sidewalls shall be without backs. The double row benches installed along the centerline of the semitrailer

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

shall utilize a vertical common single-board thickness back not less than 22 inches (559 mm) high. The oakwood slats shall be fitted between vertical supports with rounded smooth surfaces. The bench depth shall be not less than 1 foot (305 mm). Benches shall be arranged as follows:

a. In the kickup for classes A and B: One row along each side and a double row, back to back, along the centerline of the semitrailer for the full length of the kickup.

b. In the drop section for class A: One row along each side beginning not less than 1-foot (305 mm) back from the kickup on the left (street) side and not less than 1-foot (305 mm) back from the door opening on the right (curb) side and shall extend to the rear of the semitrailer. The benches at the rear of the semitrailer shall be widened to completely cover the wheelhousings. When specified (see 6.2), a double row of seats, back to back, shall be provided on drop section centerline, beginning at the rear of the door opening and extending to 5 feet $\pm$ 6 inches (1 524 mm $\pm$ 152 mm) from rear of the semitrailer.

c. In the drop section for class B: A full row of benches shall extend to the rear of the semitrailer on the left (street) side, beginning not less than 1-foot (305 mm) back from the kickup. On the right (curb) side, a row of benches shall extend from not less than 1-foot (305 mm) from the front door openings to the front of the rear door opening and from not less than 1-foot (305 mm) back of the rear of the semitrailer. On the centerline of the semitrailer, beginning at the rear of the front door entrance and continuing to the front of the rear door entrance, there shall be a double row of benches installed back to back. The benches over the wheelhousings shall be widened to completely cover the housings.

3.8.2.9 Stanchions and rails. All stanchions, grabrails, guardrails, and grabrail supports shall be 0.1046-inch (2.7 mm) thickness tubing of stainless steel, stainless class steel, or plastic coated steel; shall not be less than 1.25-inch (32 mm) outside diameter, except the vertical stanchions in the step wells shall not be less than 1.5-inch (38 mm) outside diameter. Stanchions and rails shall be provided in the semitrailer body for passenger support as follows:

a. In the kickup: Stanchions, one adjacent to the aisle corner of each bench, shall be attached to the semitrailer floor and ceiling and shall be located 2 inches (51 mm) forward of the kickup. Three guardrails, one from each side of the semitrailer to the nearest stanchion, and one connecting the stanchions adjacent to the center benches, shall be installed. The longitudinal axis of the guardrails shall be not less than 2.5 feet (1 524 mm) from the floor.

b. In the drop section: Three vertical stanchions shall be attached to the floor and to the ceiling and located along the longitudinal centerline of the semitrailer, one at the rear approximately 6 inches (152 mm) forward of the rear emergency exit, one approximately 58 inches (1 473 mm) forward of the rear emergency exit, and one at the front approximately 46 inches (1 168 mm) rearward of the face of the kickup.

   Two vertical stanchions shall be located at the rear approximately 6 inches (152 mm) forward of the rear emergency exit, one each side of and approximately 24 inches

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

(610 mm) from the longitudinal centerline of the semitrailer, and attached to the floor and to the ceiling. One vertical stanchion shall be located in the step well, in the center of the curbside opening and attached to the rear of the bottom step and to the ceiling. Two longitudinal horizontal grabrails, 75 inches (1 905 mm) from the floor, one on each side of the semitrailer approximately 24 inches (610 mm) from the longitudinal centerline shall each extend from the rear stanchion on its respective side, forward to a point approximately 46 inches (1 168 mm) rearward of the face of the kickup, and each supported at that point by a vertical support attached to the ceiling and a horizontal support attached to the sidewall. The two grab rails shall also be sufficiently supported by an additional number of properly spaced vertical supports attached to the ceiling and horizontal supports attached to the respective sidewalls. Two horizontal grabrails, both located on the longitudinal centerline of the semitrailer at the rear, one 75 inches (1 905 mm) above the floor and one 42 inches (1 067 mm) above the floor, shall extend from the rear stanchion to the second stanchion from the rear. One horizontal grabrail located on the longitudinal centerline of the semitrailer at the front of the drop section, 75 inches (1 905 mm) above the floor shall extend from the foremost centerline stanchion, forward to a point approximately 24 inches (610 mm) rearward of the face of the kickup. Two additional horizontal grabrails shall extend from the forward ends of the side grabrails to the center grabrails at its forward end. The three grabrails shall be supported at their point of intersection by a vertical support attached to the ceiling.

c. When possible, all stanchions, grabrails, and grabrail supports shall be attached to sidewall posts or roof bows as applicable. When not possible to attach to sidewall posts or to roof bows because of spacing, reinforcement shall be welded between the sidewall posts or the roof bows, stanchions, grabrails, and supports attached to the reinforced members. All stanchions, rails, and supports shall be provided with flanges which shall be secured as above with not less than two cadmium plated, self-tapping, metal screws of not less than 0.25-inch (6 mm) diameter in size.

3.8.2.10 Start and stop signal. Signal shall consist of weatherproof air actuated horn of adequate size and volume to signal operator of towing vehicle of desired stops and starts and shall be mounted outside the semitrailer at the forward end. Controls for operation of the horn shall be located adjacent to the door control.

3.8.2.11 Dome lights. Five 12V dome lights shall be provided in the semitrailer; one mid-way in the raised section, one over kickup, one 2 feet (610 mm) forward of the emergency exit, and two equally spaced between the foregoing. Control switches for operation of the lights shall be located adjacent to the door control.

3.8.2.12 Marking. In addition to the marking specified in 3.2.1.1, the marking below shall be located on the forward portion of the trailer on both street and curbsides. Letters shall be 1.5-inch (38 mm) in height, manufactured of nonreflectorized tape in a style as specified in MIL-STD-1223.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

"ON STATION USE ONLY
30 MPH (48 KM/H) MAX"

3.8.2.13 Highway warning kit. Type II (personnel), semitrailer shall be furnished with three emergency reflective triangles conforming to DoT Federal Motor Carrier Safety Regulation 393.95(f)(2)(i), and stowed in an area that is visible and readily accessible to the passenger(s).

3.8.2.14 Fire extinguisher. Type II (personnel), semitrailer shall be furnished with three class I, size 5 fire extinguishers with mounting brackets. The fire extinguishers shall be in accordance with A-A-393. The fire extinguishers shall be mounted in an area that is visible and readily accessible to the passengers.

3.8.3 Type III (refrigerator). Refrigerator semitrailer shall be fully enclosed and insulated, and shall have a continuous straight floor without wheelhousing or kickup. The interior and exterior panels shall be straight with no bulges resulting from construction or addition of insulation. Double rear doors shall provide full width opening for access into body. Semitrailer payload capacity and clearance dimensions shall conform to 3.4.2 and 3.4.4. The overall length dimension, including door hardware and rear bumper, excluding refrigerating unit, shall be as indicated by the applicable size. The inside body dimensions shall be as follows:

a. Width, not less than 86 inches (2 184 mm).

b. Height, not less than 88 inches (2 235 mm).

3.8.3.1 Refrigeration performance. The refrigeration unit shall have sufficient cooling capacity to reduce the air temperature inside the van body, from + 100 °F (38 °C) to not more than 0 °F (-18 °C) in not more than 8 hours, while the semitrailer is in an ambient air temperature of not less than 100 °F (38 °C). The refrigeration unit shall maintain 0 °F (-18 °C) inside the van body under all normal operating conditions, including road speeds of 55 mph (88 km/h), in any ambient air temperature from 0 °F (-18 °C) to not less than + 100 °F (38 °C). The interior temperature shall be thermostatically controlled by means of a calibrated, manually adjustable device, and the refrigeration unit shall maintain, within $\pm$ 6 °F (3 °C), any selected temperature from + 35 °F (2 °C) to + 0 °F (-18 °C) when the ambient air temperature is between + 35 °F (38 °C) and + 100 °F (38 °C). The temperatures throughout the van body interior from one point to any other point shall not vary more than 2.5 °F (1.4 °C). When specified (see 6.2), the refrigeration unit shall be capable of maintaining the van body interior at 35 °F (2 °C) when the ambient air temperature is -25 °F (-32 °C) while semitrailer is operating at a road speed of 55 mph (88 km/h).

3.8.3.2 Type III insulation. Body insulation shall be nonhygroscopic, self-extinguishing, cellular material and shall be resistant to growth of fungus and retention of vermin. The thermal conductivity shall be not greater than 0.15 Btu inch per hour square foot °F (0.2 W/(m·K)) at a mean temperature of 70 °F (21 °C). Insulation shall be sealed and bonded at all joints and voids. Insulation shall be anchored to prevent sag or insulation voids.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

3.8.3.3 Inner panels. Unless otherwise specified (see 6.2), inner panels shall be of either fiberglass or exterior grade AC plywood not less than 0.25-inch (6 mm) thick with clear sealer coat on cargo side. Fiberglass panels shall be not less than 0.100-inch (2.5 mm) thick with a tensile strength of not less than 12,000 pound-force per square inch (psi) (82 728 kilopascal (kPa)). Panels shall be attached to wall structural members with fasteners spaced on not more than 2-foot (610 mm) centers. Unless otherwise specified (see 6.2), the inside area of walls and doors shall be covered from the floor to a height of not less than 2 feet (610 mm) above the floor with not less than 0.125-inch (3 mm) thick fiberglass secured with cadmium plated screws.

3.8.3.4 Type III platform. Platform shall be of extruded aluminum alloy having longitudinal corrugated or T-shaped sections not less than 2 inches (51 mm) in height and spaced on centers not greater than 2.5 inches (64 mm). The platform shall be capable of withstanding concentrated loads imposed by forklifts having 14,000 lb (6 342 kg) axle load supported by 7-inch (178 mm) wide tires spaced 30 inches (762 mm) apart on the tire centerline. The sections, with corrugations lengthwise of body, shall be welded together to prevent leakage of moisture through the floor. Sides and front of floor shall have a metal flashing extending upwards for a distance of not less than 3 inches (76 mm) and shall be sealed between metal and interior insulation to minimize sidewall condensation. Floor design shall permit semitrailer operation at speeds up to and including 55 mph (88 km/h) over highways with rated payload. Floor insulation shall be placed over body crossmembers and shall be sealed from moisture by the under pan. An under pan of 0.040-inch (1 mm) thick aluminum, 0.25-inch (6 mm) thick exterior grade AC plywood, 0.0598-inch (1.5 mm) thick steel, or 0.0239-inch (0.6 mm) thick galvanized steel on plywood shall be furnished. Not less than four self-sealing water drains shall be provided, two at front and two at rear of the platform.

3.8.3.5 Type III rear doors. Two doors, full width of the semitrailer and not less than 83 inches (2 108 mm) in height, shall provide a clear opening for access into body at rear. Doors shall be insulated so that body insulating efficiency will not be reduced. Door gasket shall be installed to ensure a positive seal. Double cam latches, with operating handle, shall be installed on each door to provide positive closing action. Door locking device shall have provisions for use of a padlock. Door shall be equipped with an inside handle operated lock release and shall be capable of being opened from inside when padlocked on the outside.

3.8.3.6 Side doors. When specified (see 6.2), the body shall be equipped with either one or two curbside doors. Door(s) shall be not less than 73 inches (1 854 mm) in height and shall provide not less than a 30-inch (762 mm) clear width opening. Door(s) shall be insulated so that body insulating efficiency will not be reduced, and furnished with gaskets to ensure a positive seal. Door locking device, with operating handle and provision for use of padlock, shall be installed on each door furnished. Door(s) shall have provision for securing in the open position.

3.8.3.7 Stiff leg supports. When specified (see 6.2), refrigerator semitrailer shall be provided with two stiff leg supports mounted at the front end. Stiff leg supports shall be of the pivot or telescoping type. The stiff legs shall pivot or extend from travel position to support position and shall be retained in place with locking pins. Legs shall be adjustable for length.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

3.8.3.8 Interior lights. Two dome lights shall be provided to illuminate the interior of the body. The lighting control switch shall be mounted inside the body near the rear door. Lighting fixtures shall be moistureproof and corrosion resistant and shall be provided with glass, plastic, or a combination of both, lenses. Lamps shall be 12V potential wired to operate from the refrigerating unit batteries.

3.8.3.9 Refrigeration unit. The refrigeration unit shall contain all components, equipment, and accessories normally furnished on the manufacturer's standard unit for comparable commercial applications and shall include the components specified herein. The refrigerator capacity shall be equal to or exceed the refrigeration performance requirements of 3.8.3.1. The refrigeration unit shall be either the single unit type or the split unit type. When a dual compartment refrigerator van is specified, all exterior thermometers mounted on the side should be either recessed or have a guard over them. The single unit type shall have all components and controls mounted at the front of the body exterior. The split unit may be either of two configurations:

a. Refrigerator unit (evaporator, condenser, and compressor) in a single compartment installed in the nose of the semitrailer and a separate engine-generator power unit mounted under the frame, or,

b. Evaporator section mounted in the nose of the semitrailer and a separate engine-condensing unit mounted under the frame.

Applicable components of the refrigeration unit shall be mounted on the semitrailer frame attached to the front of the van exterior. The unit shall incorporate lifting eyes to facilitate removal or installation. The unit shall be enclosed with either quickly removable or hinged metal panels capable of protecting the refrigeration unit under any precipitation conditions, of up to 3 inches (76 mm) of rain per hour impinging at any angle from the vertical to 45 degrees off vertical toward the unit. The unit shall extend into, but occupy a minimum of cubage within the van. The opening shall be reinforced and tightly sealed. All components and assemblies shall withstand vibration encountered in normal vehicle operation. A currently Environmental Protection Agency approved refrigerant shall be used in the unit. The refrigeration unit shall be furnished with a gasoline or diesel engine, and with an auxiliary electric motor. Means shall be provided for automatically switching the drive from the engine to the electric motor as electrical power is switched on by means of a manually operated switch, leaving the engine disconnected. The electric motor shall conform to NEMA MG1 and shall be rated for 208V, 3-phase, 4-wire, 60-cycle, alternating current (ac) operation. The refrigeration unit shall include components and controls as follows:

a. Engine choke.

b. Fuel tank, integrally or separately mounted, for 12-hour operation per filling; and fuel gage, visible from outside unit enclosure.

c. Starting motor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

d. Generator or alternator. Storage batteries and battery container, integral or separately mounted.

e. Pressure unloading device on compressor or centrifugal engine clutch.

f. Automatic defrosting effected by reverse cycle, electric, or bypassing hot refrigerant from discharge line to built-in heater in evaporator.

g. A replaceable liquid line refrigerant filter/dehydrator using renewable dehydrator.

h. Thermostatic expansion valve in liquid line.

i. Heat exchanger in suction line.

j. Digital or dial thermometer to indicate body interior temperature, readable at 5-foot (1 524 mm) distance outside the unit.

k. Refrigerant compound suction-pressure gage facing outside the unit enclosure.

l. Manually set thermostatic controls for maintaining the body interior at the temperature indicated in 3.8.3.1.

m. The unit shall have a manually operated control switch which will shut down the gasoline or diesel engine and start the electric motor. This switch shall prevent the gasoline or diesel engine from being accidentally started while the electric motor is in operation when the electric motor is connected to an outside power source.

n. Controls shall include an automatic start and stop operation or be of a modulating type.

3.8.3.10 Receptacle and caution plate. A weather-resistant slave receptacle shall be provided on the refrigeration unit in an accessible and convenient location for connection to a 208V, 3-phase, 4-wire, 60-cycle power source. Wiring shall be in accordance with NFPA 70. The receptacle shall be 3-wire, 4-pole, polarized type with female contacts. The receptacle shall be treated for use with 250V ac and of adequate capacity for the motor furnished. A mating male plug shall be provided. The receptacle shall have a spring hinged cover, except when receptacle is facing in a downward direction. Aluminum or corrosion-resistant plate with stamped letters, that will be legible in the event of being painted over, shall be furnished and mounted in a location near the slave receptacle. The plate shall be not less than 2.5 by 4 inches (63 by 102 mm) and bear the following inscription in characters of easily read size:

"Power source required: 208V, 4-wire, 60-cycles, ac.
Notice: Power source of 110V or 220V, single-phase,
will not operate the unit."

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

3.8.3.11 Air leakage. The air leakage of the trailer equipped with the refrigeration system shall not greater than 12 cubic feet per minute (cfm) (5.7 litre per second (L/s)) at an air pressure differential of 0.25 inches (6 mm) of water gage without external air movement.

3.8.3.12 Operating instructions. Operating instructions shall be provided on plate(s) or decal(s) mounted inside a removable panel of the unit enclosure. The outside of the panel shall note the location of the instructions. The operating instructions shall describe any special or important procedure to be followed in operating and servicing the unit. Plates shall be permanently affixed to the panel with screws or bolts of not less than 0.125-inch (3 mm) diameter. Printed decals shall have a pressure sensitive adhesive backing.

3.8.4 Type IV (closed van). Closed van semitrailer shall be fully enclosed and have a continuous straight floor without wheelhousings or kickup. Semitrailer payload capacity and clearance dimensions shall conform to 3.4.2 and 3.4.4. The overall length dimension, including door hardware and rear bumper, shall be as indicated by the applicable size. The inside body dimensions shall be as follows:

a. Width, not less than 92 inches (2 337 mm).

b. Height, not less than 94 inches (2 388 mm).

3.8.4.1 Interior panels. Interior panels shall be not less than 0.25-inch (6 mm) exterior grade AD plywood attached to wall structural members with fasteners spaced on not more than 1-foot (305 mm) centers. When specified (see 6.2), the inside areas of walls and door shall be covered to a height of not less than 2 feet (610 mm) above the floor with 0.0598-inch (1.5 mm) (U.S. revised standard gage No. 16) thick sheet steel. Sheets shall be treated and painted before assembly and secured in place by manufacturer's standard fastening method.

3.8.4.2 Type IV platform. The platform floor shall be of either composite or laminated hardwood type. The platform shall be capable of withstanding concentrated loads imposed by forklifts having a 14,000 lb (6 342 kg) axle load supported by 7-inch (178 mm) wide tires spaced 30 inches (762 mm) apart on the tire centerline. Composite floors shall have hardwood floorboards not less than 1.25-inch (32 mm) thick finished dimensions and not less than 6 feet (1 829 mm) long, accurately fitted and laid lengthwise between hat-shaped channels constructed of not less than 0.0747-inch (2 mm) (U.S. revised standard gage No. 14) thick steel. Floorboards shall be countersunk for, and secured with phosphate-coated or other corrosion-resistant countersunk head screws, two at each cross sill, spaced on centers of not more than 12 inches (305 mm), and two on each side of each butt joint. Butt joints shall be centered over crossmembers and staggered. Laminated hardwood floors shall be not less than 1.0625-inch (27 mm) thick finished dimension, supported on cross sills spaced on centers of not more than 12 inches (305 mm) or 1.375-inch (35 mm) finished dimension supported on cross sills spaced on centers of not more than 16 inches (152 mm). Rub rails shall extend the full length of the sides of the platform.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

3.8.4.3 Type IV doors. Double rear doors shall be provided. The clear opening shall be not less than 88 inches (2 235 mm) in width and not less than 88 inches (2 235 mm) in height. Doors shall be hung with not less than three heavy-duty strap pin hinges on each door. Doors shall permit folding back against the body sides and be provided with means for holding in the full open position. Doors shall be provided with gaskets and be tightly fitted to ensure a seal against the entrance of water and dust. Each door shall be provided with double cam latches with operating handle and have provisions for locking with a padlock. When specified (see 6.2), a rollup rear door assembly shall be provided with necessary gaskets and be tightly fitted to ensure a seal against entrance of water and dust. The door handle and device shall have a provision for locking with a padlock.

3.8.4.4 Side door. When specified (see 6.2), curbside swing type door, not less than 60 inches (1 524 mm) wide by 72 inches (1 829 mm) high, shall be furnished. The door shall be located near the center of the van body. The door shall be of aluminum faced plywood or of all aluminum with bracing to prevent sagging. The door shall be equipped with not less than three hinges. A cam-type locking device shall be provided with an operating handle and shall be installed to ensure positive closing under all operating conditions. Provisions shall be incorporated for a padlock. Handles shall be located to permit operation from ground level. The hardware shall be the manufacturer's standard. The door shall have a weather-tight rubber or plastic seal all around. The side wall shall be reinforced at the door opening. The body shall be equipped with latches to hold the door in a fully open position.

3.8.4.5 Vent door. The trailer shall be equipped with four vent doors. Two doors located on the front wall and two on the rear wall. The doors shall be not less than 6 by 12 inches (152 by 305 mm) on the front wall and not less than 12 by 12 inches (305 mm by 305 mm) on the rear wall. The doors shall have screen protectors.

3.8.4.6 Loading ramp. When specified (see 6.2), a loading ramp, not less than 14 feet (356 mm) in length shall be furnished. Ramp shall be capable of rigidly supporting a concentrated load of not less than 900 lb (408 kg) without permanent deformation. The ramp shall be stored in its own storage frame positioned between the frame rails and accessible from the rear of the trailer. The ramp for use with the side doors, shall be lightweight for easy handling when removing from and returning to storage. Unit to be similar or equal to Melcher Model Easy-14A.

3.8.4.7 Roll-up rear door. When specified (see 6.2), a roll-up overhead door shall be provided. The door shall provide a clear opening of not less than 85 inches (2 159 mm) in width and not less than 82 inches (2 083 mm) in height. Otherwise, the door shall conform to 3.8.1.6.

3.8.5 Type V (open van). Open van semitrailer shall conform to requirements of 3.8.4, except as modified herein. The inside wall height shall be not less than 88 inches (2 235 mm) and the door opening height with header installed shall be not less than 82 inches (2 159 mm). A removable door header shall be furnished. Laminated hardwood is not acceptable for flooring.

3.8.5.1 Tarpaulin bows and tiedown devices. Removable tarpaulin bows, with two longitudinal top straps or ropes, shall be furnished. Tarpaulin bows shall be of steel or aluminum, mounted in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

sockets on the cap rail, and spaced at nominal 36-inch (914 mm) intervals from the rear door header to front of semitrailer. Top straps or ropes shall be provided with anchoring devices and means for tension adjustment. Tarpaulin tiedown devices shall be installed at a height not greater than 6 feet (1 829 mm) from the ground.

3.8.5.2 Tarpaulin. When specified (see 6.2), a tarpaulin shall be provided. Canvas shall be olive drab color. Tarpaulin shall be fabricated from hard texture cotton duck cloth, not less by weight per square yard (square mm (mm^2)), than No. 8 commercial designation. The cloth shall be treated to be resistant to fire, water, weather, and mildew and all seams shall be double stitched. The tarpaulin shall cover the entire body when spread over a load level with the top of the van walls, and shall extend down the sides, front, and rear to within 4 feet (1 219 mm) of the platform. Manila tiedown ropes, 0.375-inch (10 mm) in diameter, shall be spliced through grommets spaced not more than 2 feet (610 mm) apart. The free ends of the rope shall be secured with metal clips or seized with waxed twine to prevent raveling. Tiedown ropes shall be of sufficient length to permit tying to the tiedown devices when the edges of the tarpaulin are 6 feet (1 829 mm) above the platform. Leather reinforcing pads of the size, shape, and in the position required to protect the tarpaulin at the wall corners shall be sewn to the tarpaulin.

3.8.6 Type VI (side curtain van). Side curtain van shall have a continuous straight floor without wheelhousing or kickup and shall be fully enclosed as specified herein:

a. Overall height, not more than 162 inches (4 115 mm).

b. Overall width, not more than 102 inches (2 591 mm).

c. Overall length, not more than 48 feet ±1-inch (14 630 mm ± 25 mm).

3.8.6.1 Exterior. The exterior body construction shall be aluminum alloy not less than 0.050-inch (1.3 mm) thick.

3.8.6.2 Curb and street side walls. Curb and street side walls shall be curtains in lieu of conventional aluminum sides. The curtains shall extend the full length and width of the trailer. The curtains shall ride on a curtain roller and or cable assembly. The roller assembly shall be attached to the upper roof area and shall be continuous length of the trailer. Vertical tensioning straps of not less than 1.75-inch (44 mm) wide shall be located on 24-inch (610 mm) centers. The tensioners shall have bottom hooks to secure the curtain on the under side of the frame rail. A ratchet assembly to provide a positive brake and to maintain curtain tension when closed and provide a positive release shall be provided. Opening and closing the curtain assembly shall be performed by the operator without assistance. Any additional roof supports shall be removable in order to permit easy access to the cargo loading area.

3.8.6.3 Roof panels. Roof shall be of one piece and of stretcher level aluminum not less than 0.040-inch (1 mm) thick, with roof bows on not less than 24-inch (610 mm) centers. A drip molding or extrusion with rain gutter shall be provided.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

3.8.6.4 Front wall. The front wall shall have beveled or rounded corners with a minimum sacrifice of the interior cube. The front wall shall be in accordance with DoT bulkhead requirements. The aluminum alloy skin shall be not less than 0.050-inch (1.3 mm) thick.

3.8.6.5 Interior panels. Interior panels shall be not less than 0.375-inch (10 mm) exterior grade AD plywood attached to wall structural members with fasteners spaced on not more than 12-inch (305 mm) centers.

3.8.6.6 Platform. The platform floor shall be either composite or laminated hardwood type. The platform shall be capable of withstanding concentrated loads imposed by forklifts having a 14,000 lb (6 342 kg) axle load supported by 7-inch (178 mm) wide tires spaced 30 inches (762 mm) apart on the tire centerline. Composite floors shall have hardwood floorboards not less than 1.25-inch (32 mm) thick finished dimensions and not less than 6 feet (1 829 mm) long, accurately fitted and laid lengthwise between hat-shaped channels constructed of not less than 0.0747-inch (1.9 mm) thick steel. Floorboards shall be countersunk for, and secured with phosphate-coated or other corrosion-resistant countersunk head screws, two at each cross sill, spaced on centers of not more than 12 inches (305 mm), and two on each side of each butt joint. Butt joints shall be centered over crossmembers and staggered. Laminated hardwood floors shall be not less than 1.0625-inch (27 mm) thick finished dimension, supported on cross sills spaced on centers of not more than 12 inches (305 mm) or 1.375-inch (35 mm) finished dimension supported on cross sills spaced on centers of not more than 16 inches (152 mm). Rub rails shall extend the full length of the sides of the platform.

3.8.6.7 Rear curtain door. A rear curtain shall be provided in lieu of conventional rear double doors. The curtain assembly shall be the same as the curb and street side walls (see 3.8.6.2 and 3.8.6.8). The clear opening shall be not less than 98 inches (2 489 mm). The operator shall be capable of opening the curtain without any assistance.

3.8.6.8 Curtain material. The curtain material for the side walls (see 3.8.6.2 and 3.8.6.7), shall be made of rip-stop reinforced polyester impregnated with polyvinyl chloride (PVC) material. The color of the curtain material shall be the same as the exterior trailer finish coat.

3.9 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 psi (6 894 kPa) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.10 Servicing and adjusting. Prior to acceptance of the semitrailer by the Government, the contractor shall service and adjust the semitrailer for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

b. Adjustment of brakes (when required).

c. Proper functioning of all lighting and electrical systems.

d. Wheel alignment (when required).

e. Adjustment of engine to include tune-up (see 3.8.3.9).

f. Complete lubrication with grades of lubricants recommended for ambient temperatures at the delivery point.

The semitrailer shall be conspicuously tagged to identify the lubricants and their temperature range.

3.11 Workmanship.

3.11.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.11.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.11.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. The rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.11.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.11.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or other defect which reduces the casting's ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

specified herein. Except as otherwise specified in the contract, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production trailer inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First production trailer inspection. The first production semitrailer produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. This inspection shall include the examination of 4.3 and the tests of 4.4. The purpose of the inspection shall be to determine vehicle conformity with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3 and the test of 4.5. In addition, all type III semitrailers shall be tested in accordance with paragraph 4.4.4, and in accordance with paragraph 4.4.6 to verify conformance to 3.8.3.1.

4.3 Examination. Each semitrailer shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

4.4 First production test. The first production vehicles shall be subjected to the tests specified in 4.4.1 through 4.4.12. Failure to pass any phase of these tests shall be cause for rejection.

4.4.1 Vehicle weight. The semitrailer shall be weighed to determine net weight and distribution of net weight on fifth wheel and axle(s). The imposed loading on the fifth wheel and axle(s) shall be computed using net weight and the rated payload. Calculated imposed loads fifth wheel and axle(s) shall be utilized to ascertain that tires, axles, and suspension furnished are of adequate capacity to meet the contract requirements, and to determine conformance to 3.4.5.

4.4.2 Road test. The vehicle shall be road tested for a distance of not less than 10 miles (16 km) at speeds specified in 3.5. The vehicle shall be loaded to the maximum rated payload and be driven approximately 8 miles (13 km) as specified in 3.4.2, and be driven empty for approximately 2 miles (3 km). Not less than 20 percent of the test distance loaded and twenty-five percent of the distance unloaded shall be performed over unimproved roads such as gravel or dirt. Tracking ability shall be observed to verify conformance to 3.5.2. The semitrailer shall be loaded with 1.5 times the maximum rated payload, evenly distributed in the load space, and shall stand for a period of one hour in a static position without any deformation or failure.

4.4.2.1 Test failure. The first production vehicle shall successfully complete the entire road test. Substitution of a new vehicle or replacement of a major component may require complete retest at the discretion of the Government representative. Rejection of the test vehicle shall be for damage or deficiencies, including but not limited to the following:

a. Damage caused by collision.

b. Failure of major component.

c. Vibration due to misalignment of wheels or frame.

d. Vibration due to type of construction or mounting.

e. Evidence of abnormal tire wear due to misalignment or unbalance.

f. Failure of any vehicular safety device, such as lights, electrical circuits, or brakes.

g. Evidence of structural weakness in any part of the vehicle, vehicle components, or vehicle accessories.

h. Loose mounting of parts or accessories.

i. Tracking ability does not conform to paragraph 3.5.2.

4.4.2.2 Brake test. Brake tests shall be conducted with the fully loaded semitrailer coupled to a truck-tractor. The truck-tractor used for the brake test shall be selected so that it will be loaded to full capacity (gross vehicle weight and gross combination weight) by the semitrailer being

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

tested. The truck-tractor shall be without extra brake capacity. Failure of the tractor semitrailer combination to stop within the required 40 feet (12 192 mm) from the speed of 20 mph (32 km/h) shall be cause for rejection.

4.4.2.3 Parking brake verification. The parking brakes shall be tested to show conformance to 3.6.4.2.

4.4.2.4 Landing leg verification. The landing gear shall be checked for range of adjustments as specified in 3.6.7.

4.4.3 Turning ability test. A truck-tractor shall be coupled to the semitrailer and driven through turns up to 90 degrees right and left to determine conformance to 3.5.1.

4.4.4 Leak test. The semitrailer, except type V, shall be subjected to a leak test by spraying fresh water at a rate of not less than 2 inches (51 mm) per hour at any angle of 30 degrees to 45 degrees from vertical. The test shall be performed with the water directed against all four sides and the roof of the semitrailer simultaneously, and shall continue for not less than 15 minutes. The test shall be conducted prior to the installation of the interior insulation and lining. Following the test, the semitrailer shall be inspected for water damage.

4.4.4.1 Air leakage. The type III semitrailer shall be subjected to an air leakage test. An exhaust blower, manometer, and flow meter shall be connected in series to the trailer cargo space through one floor drain. The remaining floor drains shall be sealed. Air leakage shall be measured by feeding air through a flow meter into the trailer in order to maintain 0.25-inch (6 mm) water gage pressure inside the container. The air pressure differential shall be held constant for a period of 25 minutes by throttling the exhaust blower output. Five flow meter readings shall be taken at 5-minute intervals. Nonconformance to 3.8.3.11 shall be cause for rejection.

4.4.5 Type III pulldown and operation test. The unloaded refrigerator semitrailer shall be placed in a controlled temperature room for a period of 6 hours with the temperature at 100 °F (38 °C) with the semitrailer doors open. The semitrailer doors shall be closed and the ambient temperature maintained at not less than 100 °F (38 °C). The ambient air temperatures shall be recorded at a height not more than half of the height of the semitrailer nor less than 5 feet (1 524 mm) from the ground line, while not directly exposed to the heat source. The refrigeration unit, powered by the gasoline or diesel engine, shall be operated until the temperature of the semitrailer interior is reduced to 0 °F (- 18 °C) as specified in 3.8.3.1. The period of time required to obtain this temperature shall be recorded. Immediately following this cooling test, the refrigeration unit shall maintain the interior temperature of the semitrailer at the specified low value for a period of not less than 2 hours, with equal period of operation using the engine and the electric motor. Instruments and controls of the refrigeration system shall be calibrated before, or during the test by comparison with a certified standard temperature measuring instrument calibrated in the appropriate temperature range. During the 2-hour temperature maintenance period, the temperature shall be recorded at 5-minute intervals. Temperature readings shall be made using thermocouples calibrated at the time of the test, with a digital readout instrument, preferably with a time/temperature recorder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

4.4.6 Type III cooling control test. Control of ambient air temperature will not be required provided that the ambient air temperature is maintained above +35 °F (2 °C) and below 100 °F (38 °C) during the test period. The thermostatic controls shall be set at 0 °F (- 18 °C), 20 °F (- 7 °C), and 35 °F (2 °C), and the inside body temperature brought to approximately 5 °F (- 15 °C) above each control setting. After the inside body temperature has reached each specified test value, the controls shall be tested, using the engine and the electric motor to operate the refrigeration unit, each for a period of not less than 30 minutes. The temperature shall be recorded during the test at 5-minute intervals to determine conformance to 3.8.3.1.

4.4.7 Type III highway cooling test. The thermostatic controls shall be set at + 0 °F (- 18 °C) and the semitrailer operated at the top highway speed for a period of not less than 30 minutes in an ambient air temperature between + 35 °F (2 °C) and 100 °F (38 °C) to determine conformance to 3.8.3.1. When the refrigerator unit is specified to be capable of maintaining the van body interior at 35 °F (2 °C), when ambient temperature is -25 °F (- 32 °C), and the semitrailer is traveling at 55 mph (88 km/h), proper operation of the heat cycle shall be demonstrated immediately after the highway cooling test by switching the unit to the highest heat mode to demonstrate a temperature rise on the thermometer of not less than 15 °F (- 9 °C) in 30 minutes. In lieu of the test, the contractor may submit a certification from the manufacturer that the refrigeration unit will meet the above requirements.

4.4.8 Type III rain test. The refrigeration unit shall be subjected to a simulated rain storm to determine compliance with 3.8.3.9. Fresh water shall be sprayed on the refrigeration unit at a rate of 3 inches (76 mm) per hour at any angle between vertical and 45 degrees off vertical against the exposed sides of the unit. The refrigeration unit shall be exposed to the spray for one-half hour while idle and then shall be started and operated for one-half hour while the spray continues.

4.4.9 Electrical test. An electrical test shall be conducted on the first production vehicle to determine that all vehicle lights are functioning. In addition, when an interconnected 24V system is furnished, (see 3.6.6.4), the circuits of the 12-contact receptacle shall be energized with not less than 28V regulated voltage for a time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 77 °F (25 °C). After the test, the resistor assembly, including resistors, wiring, and adjacent components shall be examined for evidence of overheating, deterioration, or damage.

4.4.10 Type IV curtain assembly test. A curtain assembly test shall be conducted to verify the conformance to 3.8.6.2 and 3.8.6.7. The test will be without assistance to the operator, the operator shall then close all the curtains and secure the assembly within 10 minutes. The total time shall be not greater than 20 minutes. Failure to pass any phase of the test shall be cause for rejection.

4.4.11 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of the vehicles under the contract until corrective action has been taken.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

4.4.12 Production sample. Upon acceptance of the first production vehicle, the first production vehicle shall remain at the manufacturing facility as a production sample, and shall be the last semitrailer delivered on the contract. The first production vehicle shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first production vehicle in a serviceable condition for the duration of the contract.

4.5 Production semitrailer test. Each semitrailer produced in fulfillment of a contract or order shall be completely assembled, adjusted, lubricated, and otherwise serviced for operation. The contractor's testing system shall, as a minimum, ensure that the semitrailer is capable of meeting the performance requirements specified herein. Failure to pass any phase of this test shall be cause for rejection.

5. PACKAGING

5.1 Packaging requirements. The preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Semitrailers covered by this specification are intended for general administrative use by the Government in transporting personnel or cargo. Type III, IV, V, and VI semitrailers may be loaded with cargo by the use of forklifts.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class and size of semitrailer required (see 1.2).
c. As specified by the procuring activity, the exterior and markings shall be in accordance with MIL-STD-1223 (see 3.2.1.1).
d. When rustproofing is required (see 3.2.1.2).
e. When lifting and tying down attachments are required (see 3.2.1.6).
f. When cargo tiedown rings are required (see 3.2.18).
g. When air suspension system is required (see 3.6.1.1).
h. When disk type wheels are required (see 3.6.3.1).
i. When wheel mounted mile meter is required (see 3.6.3.1).
j. When a spare tire carrier is required (see 3.6.3.2).
k. When a spare wheel or rim is required (see 3.6.3.3).
l. When spare tire is required (see 3.6.3.4).
m. When independent 24V lighting system is required (see 3.6.6.3).
n. When an interconnected 24V lighting system is required (see 3.6.6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

o. Body construction is other than as specified and if specific skin material is required (see 3.7).
p. Exterior skin material is specific material is required (see 3.7).
q. When sliding type door(s) are required on type I semitrailer (see 3.8.1.4).
r. When roll-up overhead rear door is required (see 3.8.1.6).
s. When roll-up overhead type door is required (see 3.8.1.7).
t. When expanded aluminum mesh, covering window openings, is required (see 3.8.2.7).
u. When centerline seats are required for type II, class A semitrailer (see 3.8.2.8 (b)).
v. When refrigerator unit shall be capable of maintaining the van body interior at 35 °F (2 °C) when ambient temperature is - 25 °F (- 32 °C) while semitrailer is operating at a road speed of 55 mph (88 km/h) (see 3.8.3.1).
w. Inner panel requirements for type III semitrailer is other than as specified (see 3.8.3.3).
x. When interior of walls and doors shall not be covered from the floor to 2 feet above the floor with fiberglass 0.125-inch thick (see 3.8.3.3).
y. When curbside door(s) are required on type III semitrailer (see 3.8.3.6).
z. When stiff leg supports are required on type III semitrailer (see 3.8.3.7).
aa. When interior panels on type IV semitrailer shall have sheet steel lining to a height of not less than 2 feet above the floor (see 3.8.4.1).
bb. When a rollup rear door assembly is required (see 3.8.4.3).
cc. When a side door for type IV semitrailer is required (see 3.8.4.4).
dd. When a loading ramp is required (see 3.8.4.6).
ee. When a roll-up overhead door is required (see 3.8.4.7)
ff. When tarpaulin is required for type V semitrailer (see 3.8.5.2).

6.3 Definitions.

6.3.1 Improved road. An improved road is a smooth, hard surfaced road, such as concrete or asphalt paved highway.

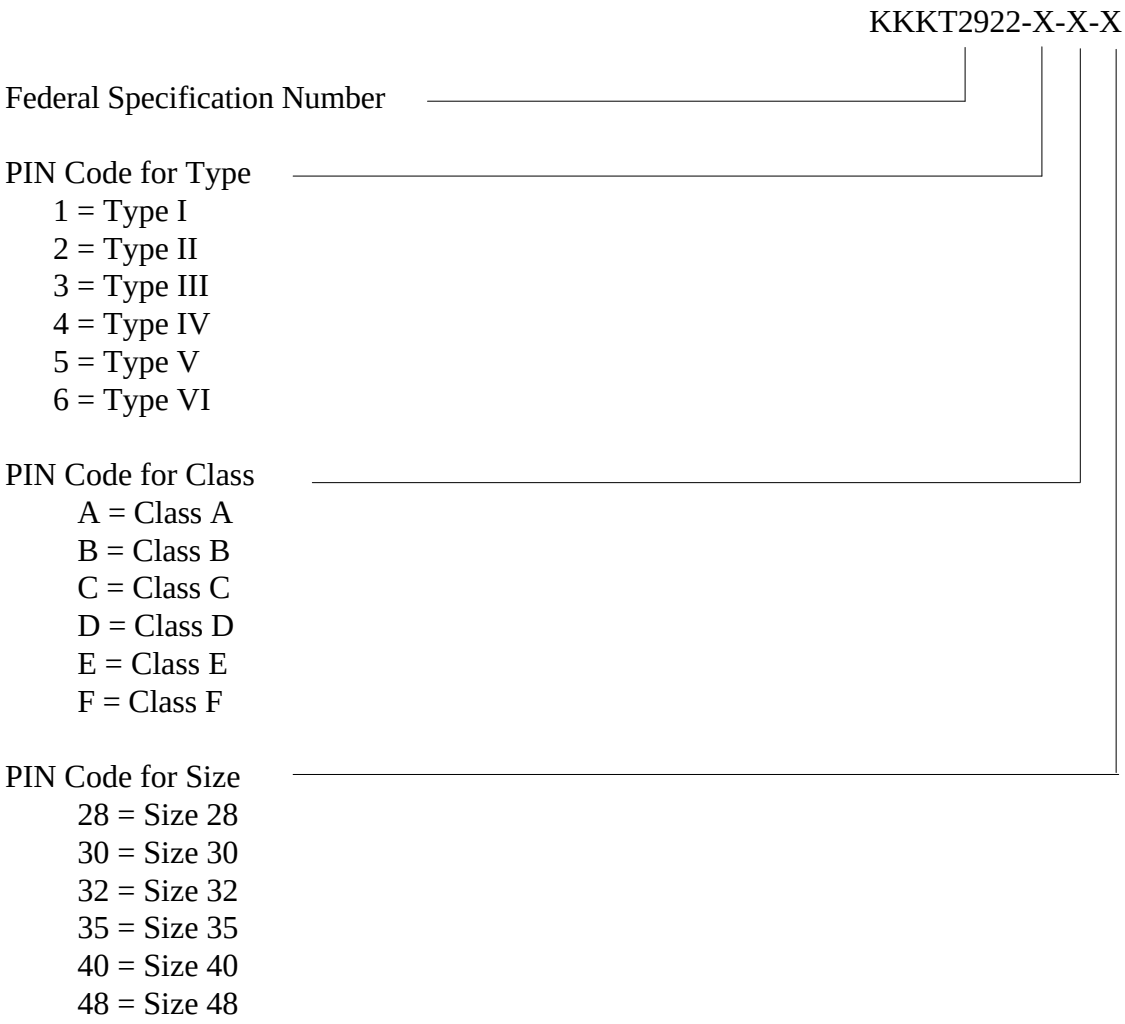
6.3.2 Unimproved road. An unimproved road is an unpaved, unstabilized road with an undulating surface having occasional chuckholes and exposed rocks.

6.4 Supersession data. This specification replaces Military Specification MIL-S-45344H dated 4 December 1991.

6.5 Classification cross reference. Classifications used in this specification (see 1.2) are identical to those found in the superseded Military Specification MIL-S-45344H.

6.6 Part or identifying number (PINs). The specification number, types, classes, and sizes are combined to form PINs for semitrailers covered by this document (see 1.2). PINs for the semitrailers are established as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.



6.7 Subject term (key word) listing.

Brake drums
Landing gear
Performance
Suspension system (vehicle)

6.8 Changes from previous issue. Marginal notations are not used to identify changes with respect to the superseded document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS:

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Custodians:
Army - AT
Navy - YD1

Review Activity:
Navy - MC

Preparing Activity:
Navy - YD1

(Project 2330-0264)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** KKK-S-2922	**2. DOCUMENT DATE (YYMMDD)** 970421

3. DOCUMENT TITLE SEMITRAILER, VAN; COMMERCIAL (FURNITURE, PERSONNEL, REFRIGERATOR, CARGO, AND SIDE CURTAIN) 7 TO 35 TON

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME R. E. PAMPLONA	b. TELEPHONE *Include Area Code)* (1) Commercial (805) 982-5843 (2) AUTOVON 551-5843
c. ADDRESS *(Include Zip Code)* COMMANDING OFFICER, NCBC, CODE 1581D 1000 23RD AVENUE PORT HUENEME, CA 93043-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:46Z
Check the source to verify that this is the current version before use.